# HOTEL REVENUE ANALYSIS

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# INTRODUCTION

- The hospitality industry generates large amounts of data from reservations, customer interactions, and pricing strategies. However, converting this data into meaningful insights requires effective analytical tools.
- Data analytics helps hotels understand booking trends, customer behavior, revenue patterns, and operational performance.
- This project focuses on analyzing hotel booking data using **Power BI**, a business intelligence tool that enables interactive dashboards and data visualization.
- By transforming raw data into dashboards, the project aims to help hotel management make **data-driven decisions**.

# PROBLEM STATEMENT

- Hotels often collect large amounts of booking data but face difficulties in extracting meaningful insights from it.
- Without proper analysis:

  Revenue performance cannot be tracked effectively.

  Customer booking behaviour remains unclear.

  Pricing strategies may not be optimized.

  Operational patterns are difficult to analyze.
- Therefore, there is a need to develop an **analytical system that converts raw hotel booking data into actionable insights.**

# PROJECT OBJECTIVES

- The primary objectives of this project are:
- To analyze hotel booking data using business intelligence techniques.
- To identify patterns in customer booking behaviour.
- To evaluate revenue performance and pricing strategies.
- To understand operational trends within hotel management.
- To develop interactive dashboards using Power BI that support decision-making.

# DATASET DESCRIPTION

- The dataset used in this project contains hotel booking information from multiple years.
- It includes various attributes that describe customer reservations, hotel characteristics, and pricing details.
- Some key attributes in the dataset include:

  Arrival Date,Hotel Type,Market Segment,Distribution Channel, Customer Country

  Average Daily Rate (ADR)

  Stay Duration (Weekend and Weekday Nights)

  Booking Status (Cancelled / Not Cancelled)

Customer Type

- These attributes allow comprehensive analysis of customer behaviour and revenue patterns.

# DATA PREPROCESSING

- Before building the dashboards, the dataset underwent several preprocessing steps to ensure accuracy.
- The preprocessing steps included:

  Data cleaning

  Handling missing values

  Removing duplicate entries

  Data transformation

  Creating calculated measures
- These steps ensured that the dataset was reliable for further analysis and visualization.

# DATA MODELING

- Data modeling is an important step in building analytical dashboards.
- In this project, the data was structured to support efficient analysis.
- The dataset was organized into:

  A central table containing booking transactions

  Supporting dimensions such as hotel type, market segment, and booking channels
- This structured data model allowed Power BI to generate fast queries and accurate insights.

# TOOLS AND TECHNOLOGIES

The following tools and technologies were used in this project:

- **Microsoft Power BI**
  Used for data preprocessing building interactive dashboards and data visualization.
- **Microsoft Excel**
  Used for initial data handling.
- **Business Intelligence Techniques**
  Applied to transform raw data into meaningful insights.

These tools helped convert complex booking data into easy-to-understand visual analytics.

# DASHBOARD ARCHITECTURE

- To analyze the dataset effectively, four dashboards were created:
- Executive Overview
- Booking Behavior Analysis
- Revenue & Pricing Analysis
- Operational Insights
- Each dashboard focuses on a different aspect of hotel performance, providing a comprehensive analysis.

# DASHBOARD 1: EXECUTIVE OVERVIEW

- **Title:** Executive Overview Dashboard
- This dashboard provides a **high-level summary of hotel performance**.
- Key performance indicators displayed include:

  Total Revenue, Total Bookings , Average Daily Rate (ADR),Occupancy Rate, Average Length of Stay.
- Visualizations included in this dashboard:

  Revenue Trend (Monthly)

  Booking Trend

  Revenue by Hotel Type

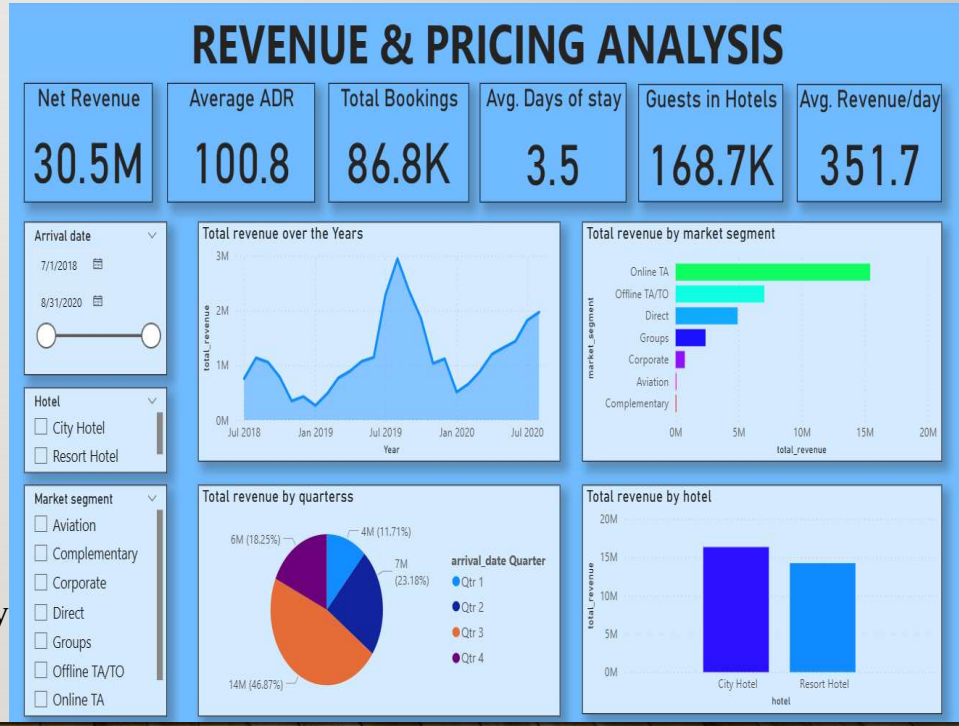
  Revenue by Market Segment
- This dashboard helps management quickly understand overall hotel performance.

# DASHBOARD 2: BOOKING BEHAVIOR ANALYSIS

- This dashboard focuses on understanding customer booking patterns.
- Visualizations included:
  - Lead Time Distribution
  - Booking by Customer Type
  - Booking Distribution by Channel
  - Cancellation Rate
  - Repeat vs New Guests
- These insights help hotels understand how customers make reservations and identify opportunities to improve customer retention.

# DASHBOARD 3: REVENUE & PRICING ANALYSIS

- **Title:** Revenue & Pricing Analysis
- This dashboard focuses on analyzing hotel pricing strategies and revenue performance.
- Key visualizations include:
  - ADR Trend
  - Revenue by Room Type
  - Revenue by Season
  - Revenue vs ADR (Scatter Plot)
  - Revenue per Booking KPI
- This dashboard helps evaluate pricing effectiveness and identify opportunities for revenue optimization.



# DASHBOARD 4: OPERATIONAL INSIGHTS

- **Title:** Operational Insights
- This dashboard provides insights related to hotel operations.
- Key visualizations include:

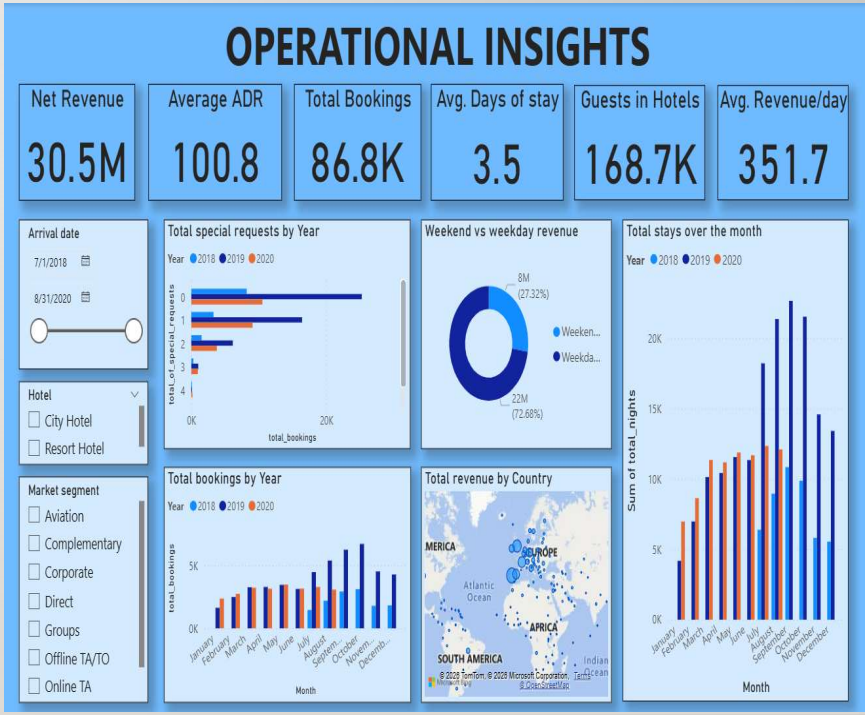
  Stay Duration Distribution

  Weekday vs Weekend Revenue

  Impact of Special Requests

  Occupancy by Month

  Top Countries by Bookings (Map Visualization)
- These insights help hotels improve operational planning and resource management.



# KEY INSIGHTS

- The analysis revealed several important insights:
- Travel agents contribute a large portion of bookings.
- The cancellation rate is relatively high.
- Most guests are new customers rather than repeat customers.
- Seasonal trends significantly impact hotel revenue.
- Certain room types generate higher revenue compared to others.
- These insights help hotels improve marketing, pricing, and operational strategies.

# CONCLUSION

- This project demonstrates how **data analytics and visualization can transform hotel booking data into valuable insights**.
- Through interactive dashboards, the analysis provides a clear understanding of:

  Revenue performance

  Customer booking behavior

  Pricing strategies

  Operational insights
- These insights help hotel management make better strategic decisions and improve overall business performance.